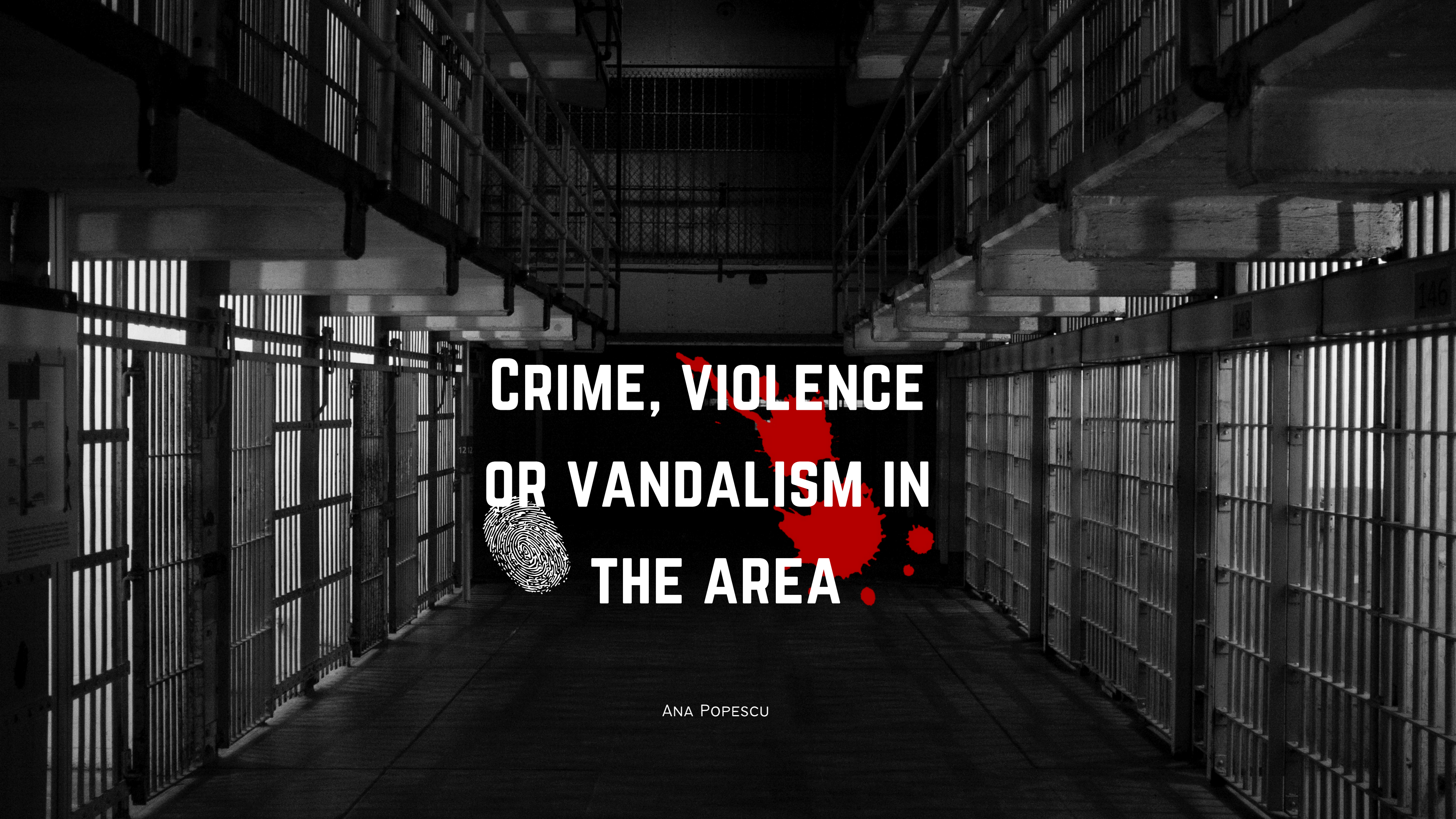




CRIME, VIOLENCE OR VANDALISM IN THE AREA

ANA POPESCU

BIBLIOGRAPHY

NUMBER OF POLICE OFFICERS

https://ec.europa.eu/eurostat/databrowser/view/crim_just_job/default/table?lang=en

NUMBER OF PRISONERS

https://ec.europa.eu/eurostat/databrowser/view/crim_pris_ctz/default/table?lang=en

SUSPECTS AND OFFENDERS

https://ec.europa.eu/eurostat/databrowser/view/crim_just_ctz/default/table?lang=en

INTENTIONAL HOMICIDE

[https://ec.europa.eu/eurostat/databrowser/view/CRIM_HOM_SOFF__custom_4363519/default/table?
lang=en](https://ec.europa.eu/eurostat/databrowser/view/CRIM_HOM_SOFF__custom_4363519/default/table?lang=en)

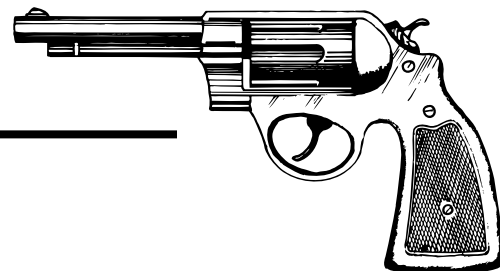
LEGAL CASES PROCESSED

[https://ec.europa.eu/eurostat/databrowser/view/CRIM_CRT_CASE/default/table?
lang=en&category=crim.crim_crt](https://ec.europa.eu/eurostat/databrowser/view/CRIM_CRT_CASE/default/table?lang=en&category=crim.crim_crt)

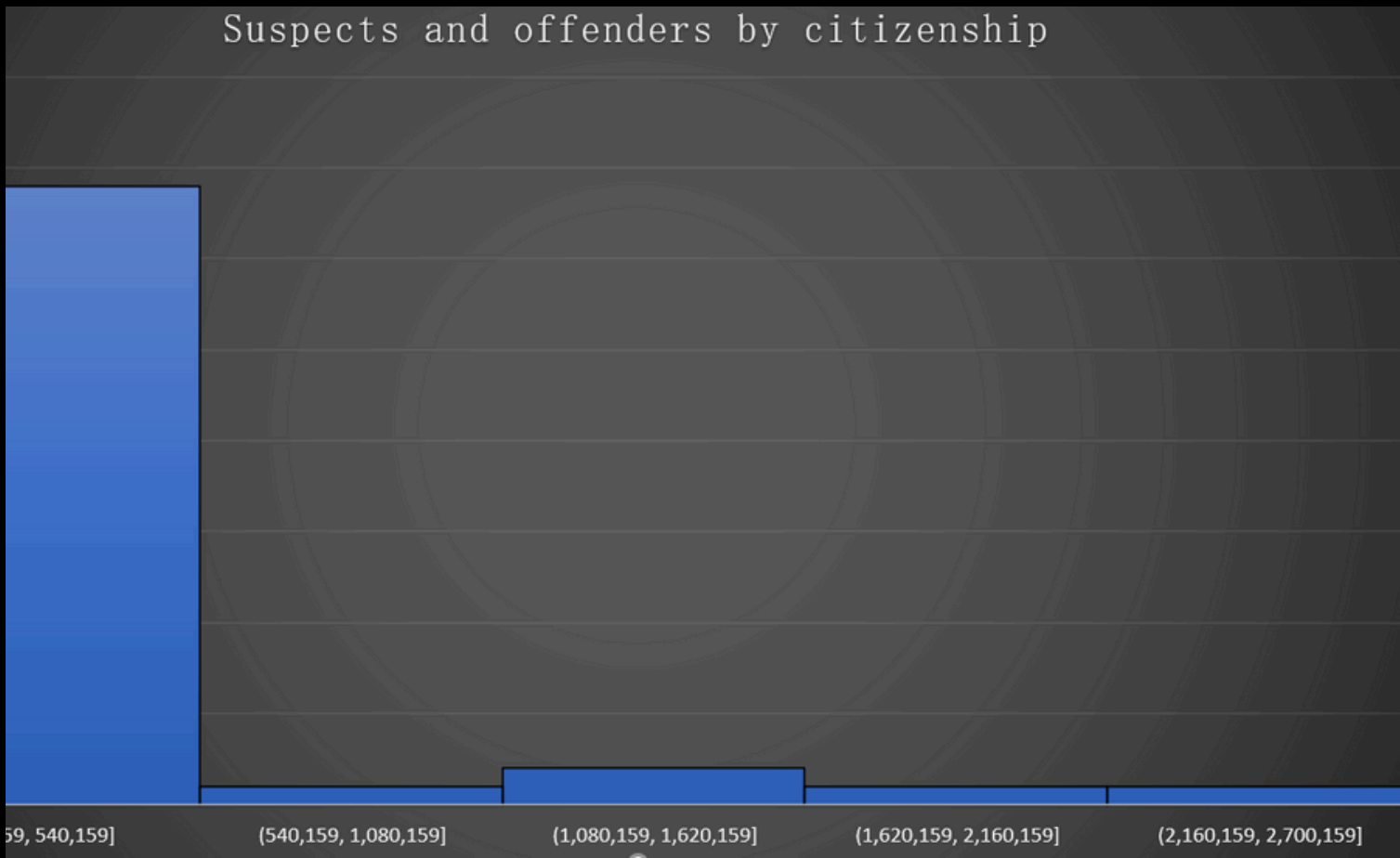
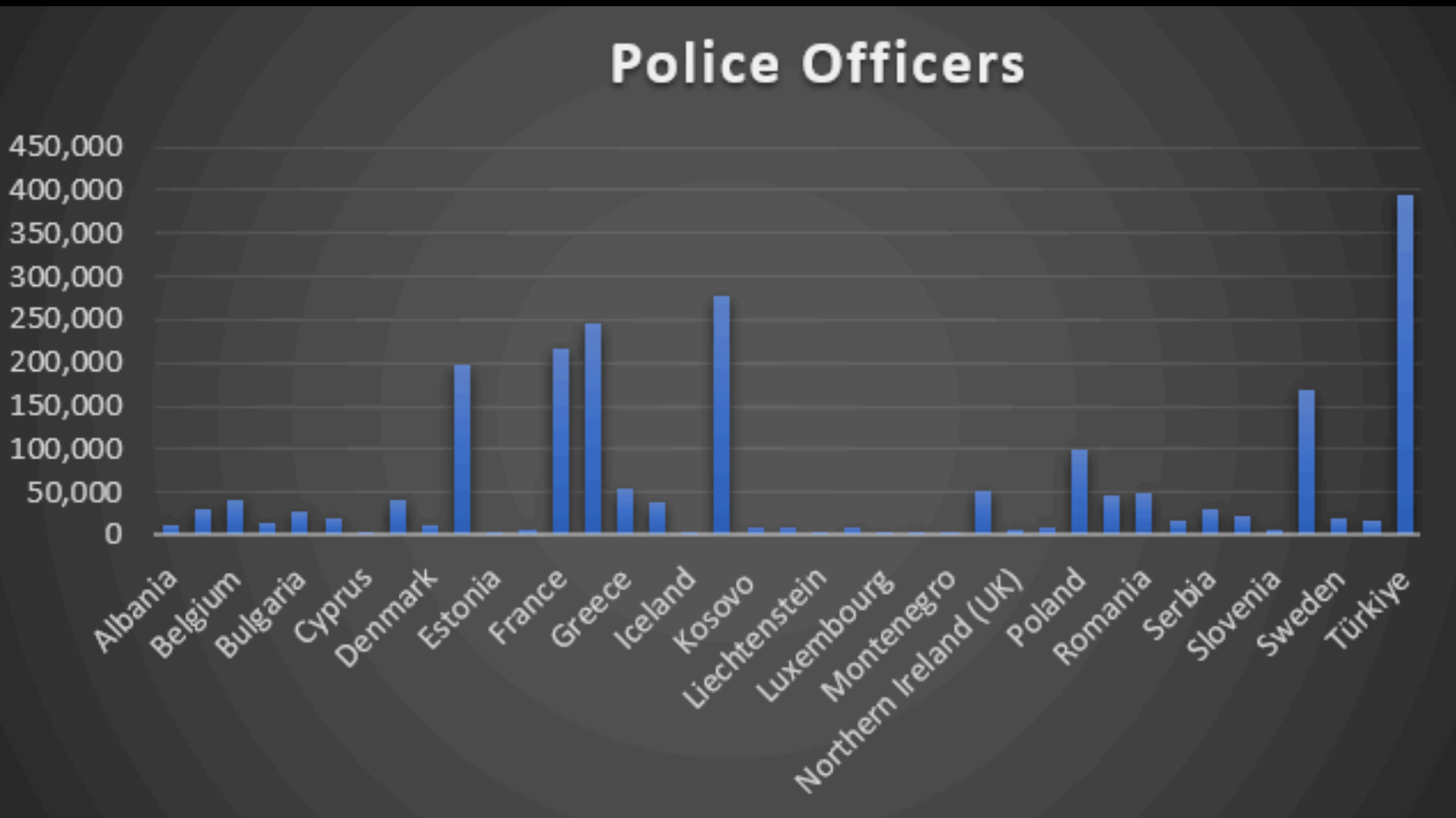
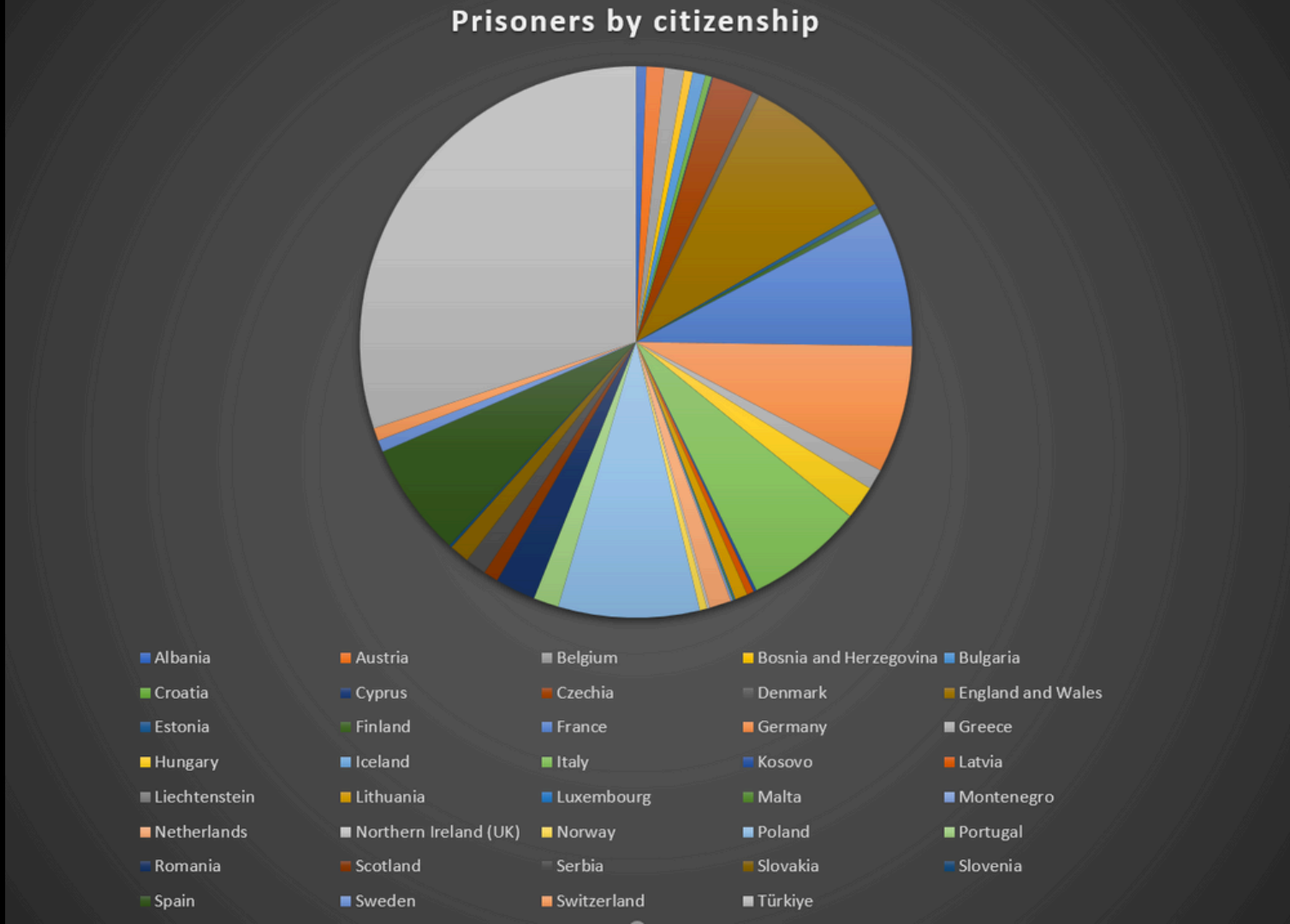
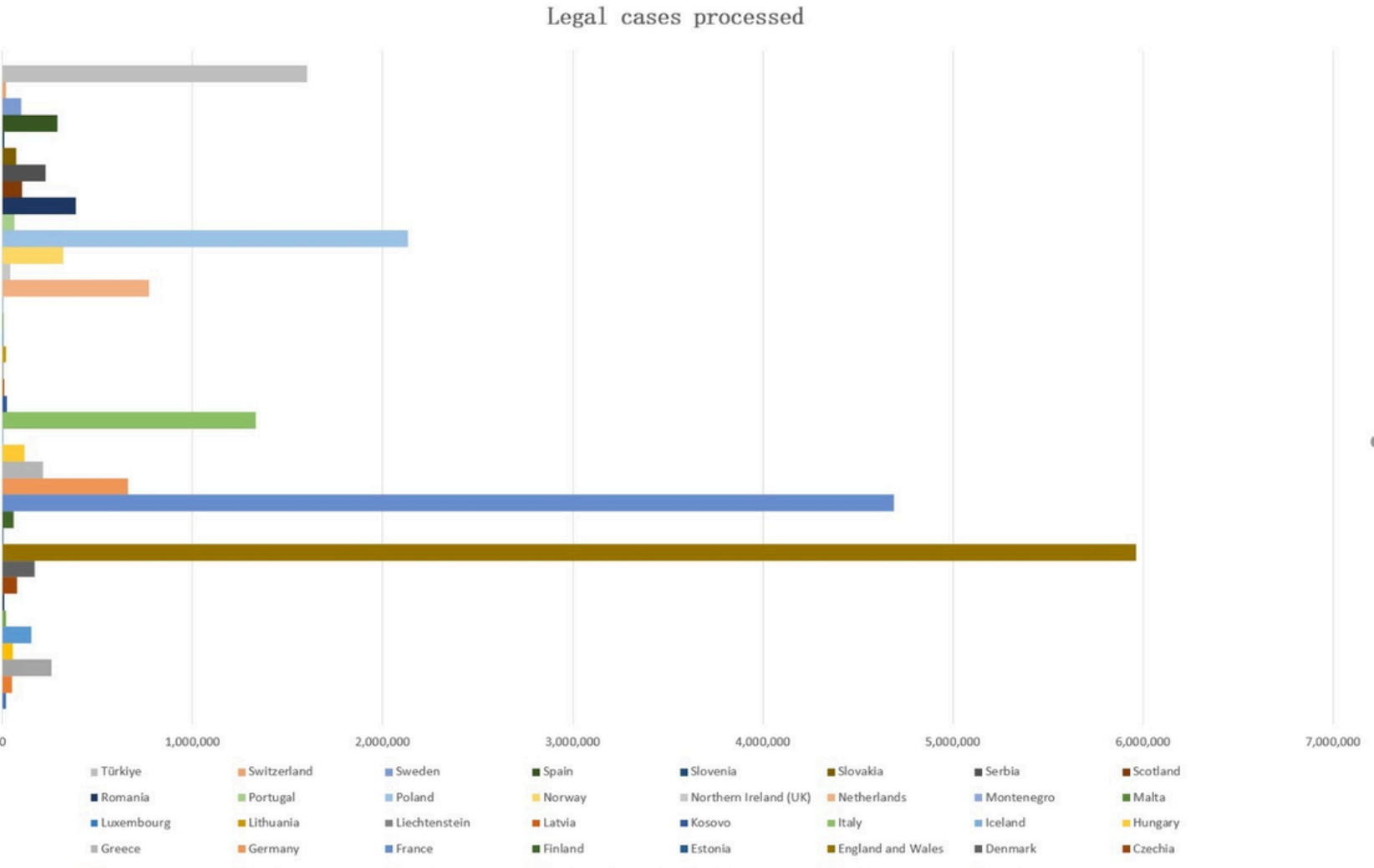
			2018			
GEO (Labels)	Legal cases processed	Prisoners by citizenship	Suspects and offenders by citizenship	Police officers	Intentional homicide(convicted)	
Albania	16,883	5,316	39,649	10,572	114	
Austria	51,160	9,163	288,414	30,240	96	
Belgium	258,500	10,261	198,634	41,370	97	
Bosnia and Herzegovina	51,872	4,377	19,216	15,192	98	
Bulgaria	150,324	6,651	48,449	28,742	72	
Croatia	17,113	3,217	17,399	20,199	54	
Cyprus	8,438	619	4,347	4,927	2	
Czechia	74,115	21,577	84,620	40,040	103	
Denmark	167,564	3,635	34,042	11,050	32	
England and Wales	5,965,003	81,904	1,190,000	198,686	726	
Estonia	5,876	2,584	75,191	3,893	28	
Finland	58,716	2,956	454,726	7,317	40	
France	4,687,990	70,059	1,178,414	215,948	315	
Germany	659,932	65,762	2,070,987	244,800	297	
Greece	214,663	10,654	124,257	53,166	94	
Hungary	115,730	17,251	52,475	39,423	86	
Iceland	1,858	136	6,596	619	89	
Italy	1,331,892	61,131	869,304	276,750	672	
Kosovo	23,640	1,567	34,450	8,633	121	
Latvia	9,164	3,522	13,384	8,049	102	
Liechtenstein	26	63	159	120	1	
Lithuania	19,746	6,485	24,075	8,247	114	
Luxembourg	4,500	656	29,245	1,982	0	
Malta	1,410	635	5,791	2,289	7	
Montenegro	3,781	1,125	4,425	4,105	6	
Netherlands	770,447	11,251	165,960	50,389	585	
Northern Ireland (UK)	42,553	1,407	20,856	6,741	16	
Norway	317,927	3,791	70,497	10,170	24	
Poland	2,134,568	72,818	336,886	98,709	173	
Portugal	61,827	13,021	174,822	46,363	76	
Romania	383,576	20,792	196,461	50,024	686	
Scotland	102,825	7,595	246,480	17,296	59	
Serbia	225,285	10,871	50,278	30,265	68	
Slovakia	72,209	10,294	41,426	21,918	36	
Slovenia	9,466	1,346	15,579	7,091	35	
Spain	287,852	58,883	389,149	169,139	160	
Sweden	97,472	6,114	180,770	20,040	170	
Switzerland	19,598	6,602	122,353	18,413	36	
Türkiye	1,601,160	264,842	2,246,537	395,522	6,724	

We chose this data set with the representative variables because they are essential indicators that influence the occurrence of crime, violence or vandalism in the local area by monitoring progress towards peace, justice and strong institutions.

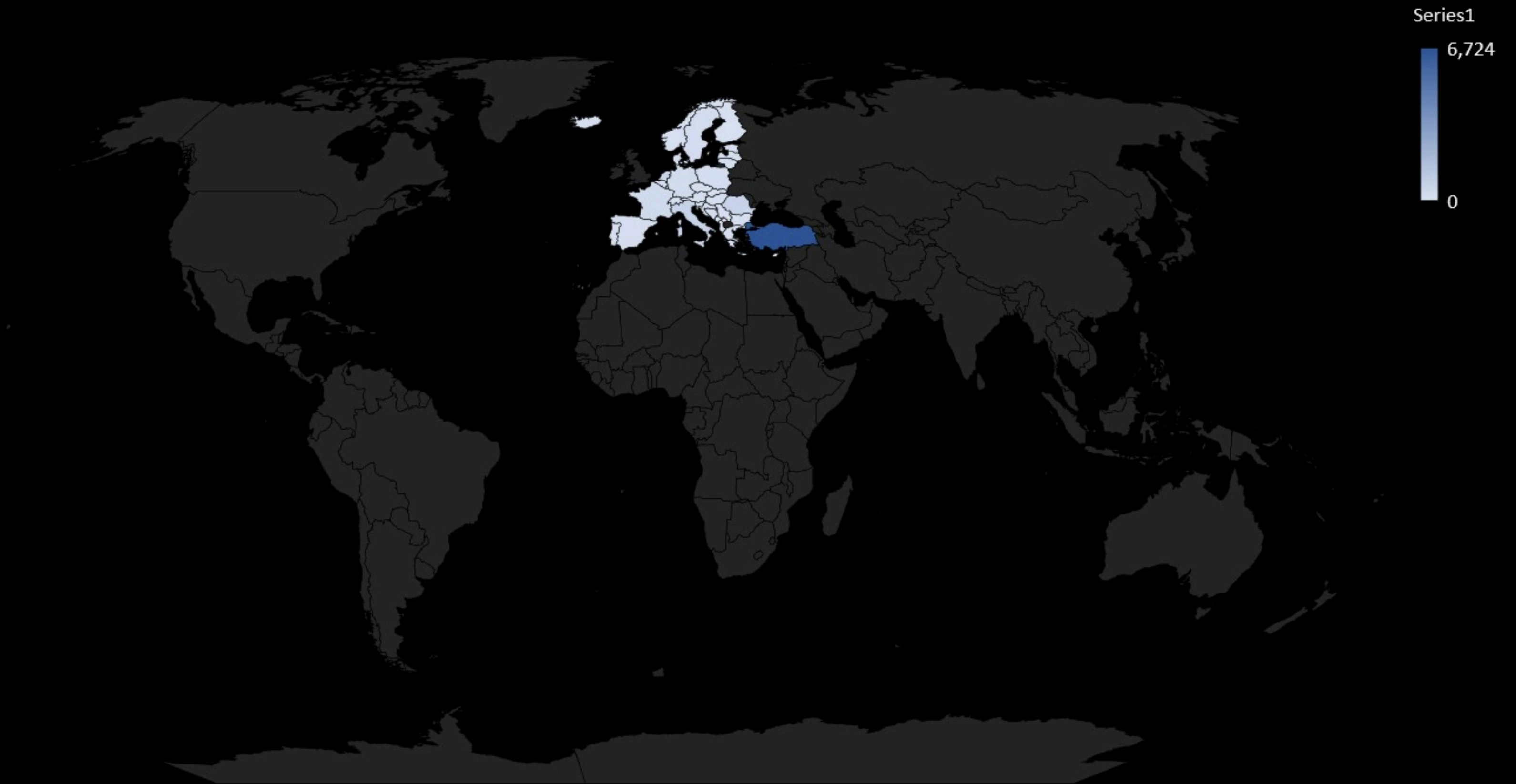
Why did we choose this data set?



As statistics confirm, 11% of the population of the 27 EU Member States reported they faced a problem of crime, violence or vandalism in their local area.



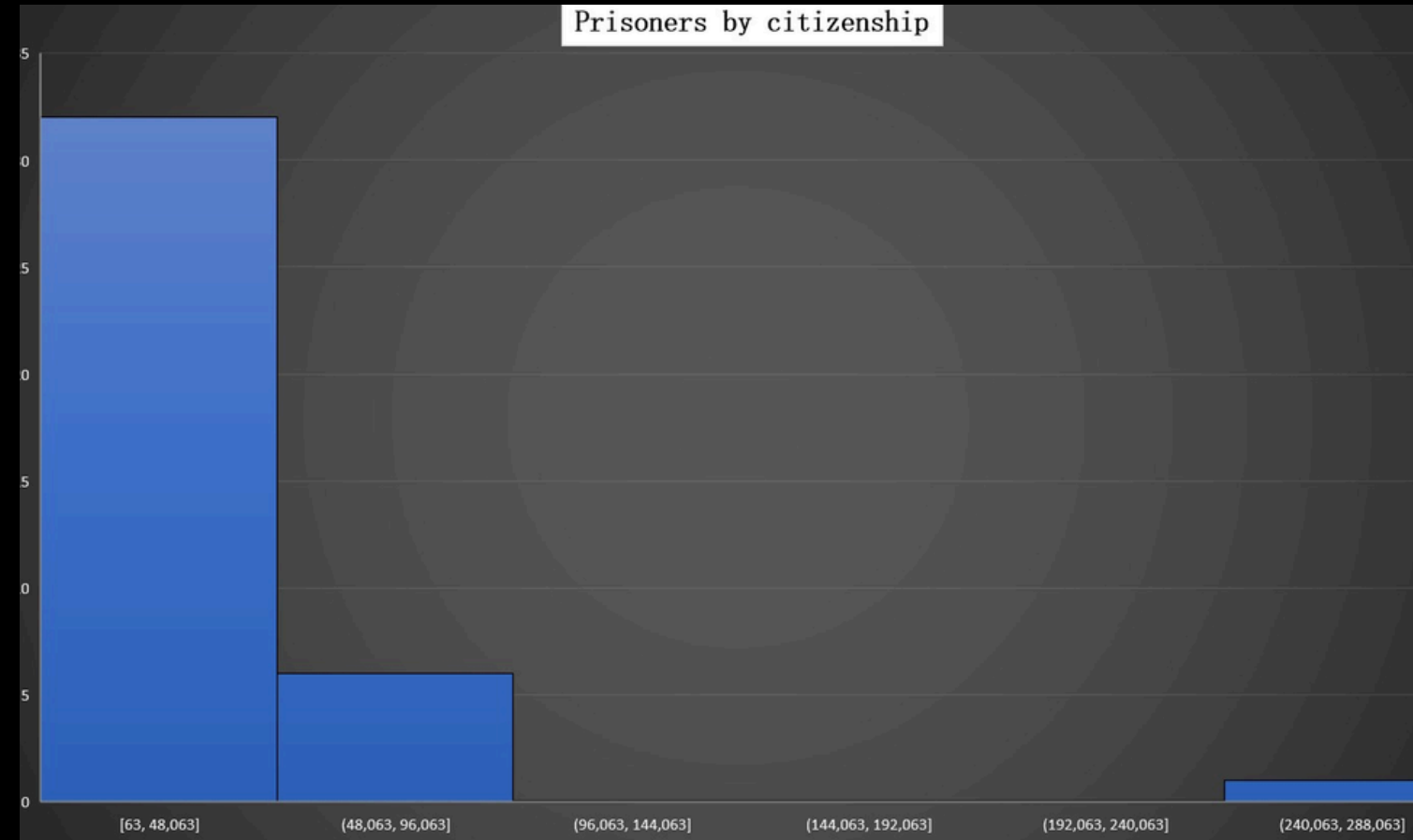
Intentional Homicide



HISTOGRAMS



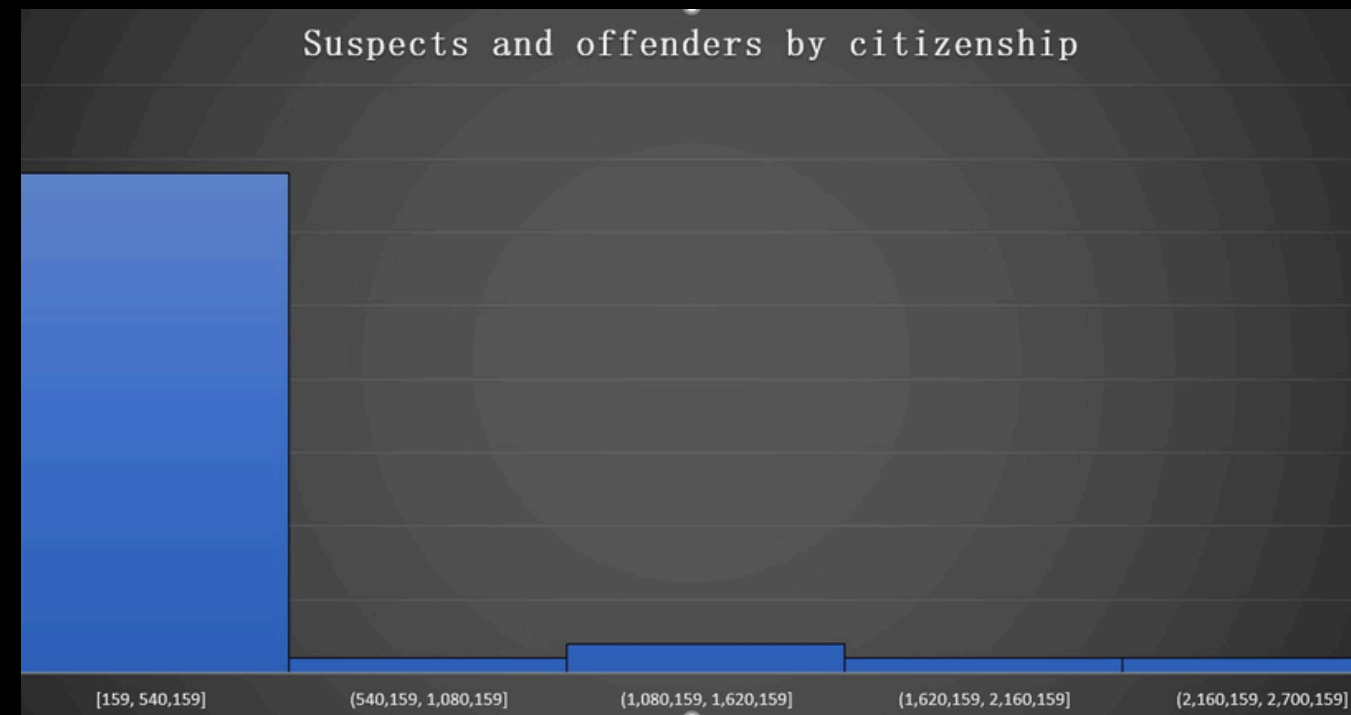
Prisoners by citizenship



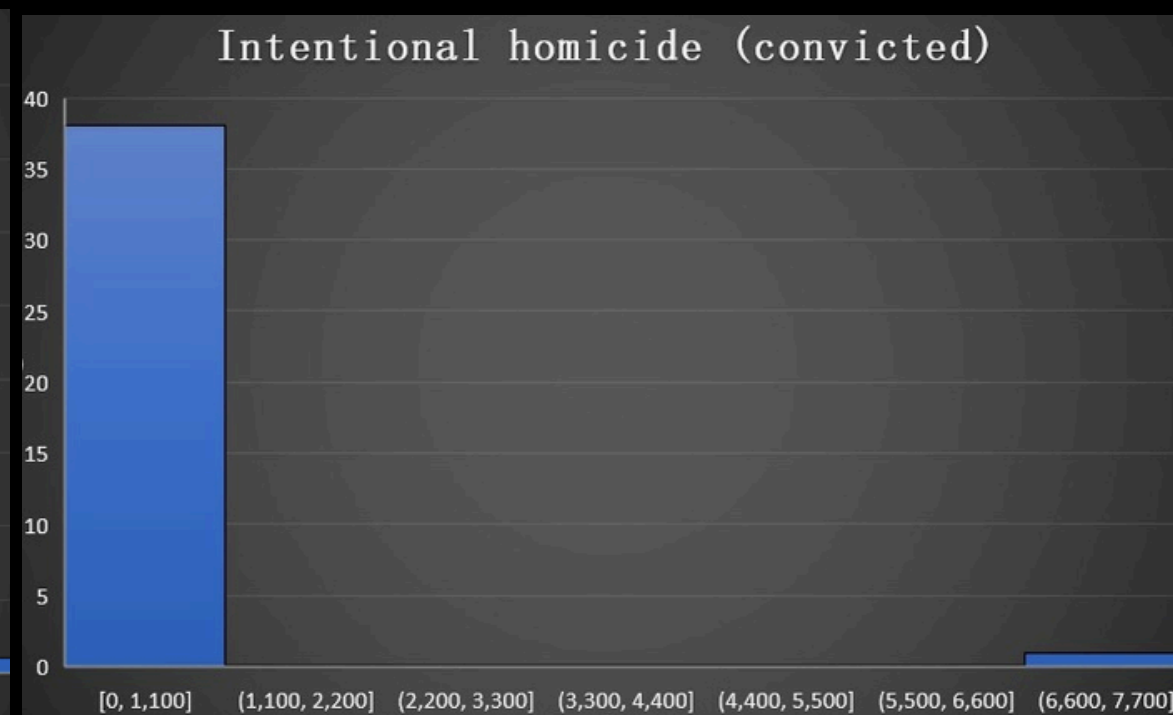
Legal cases processed



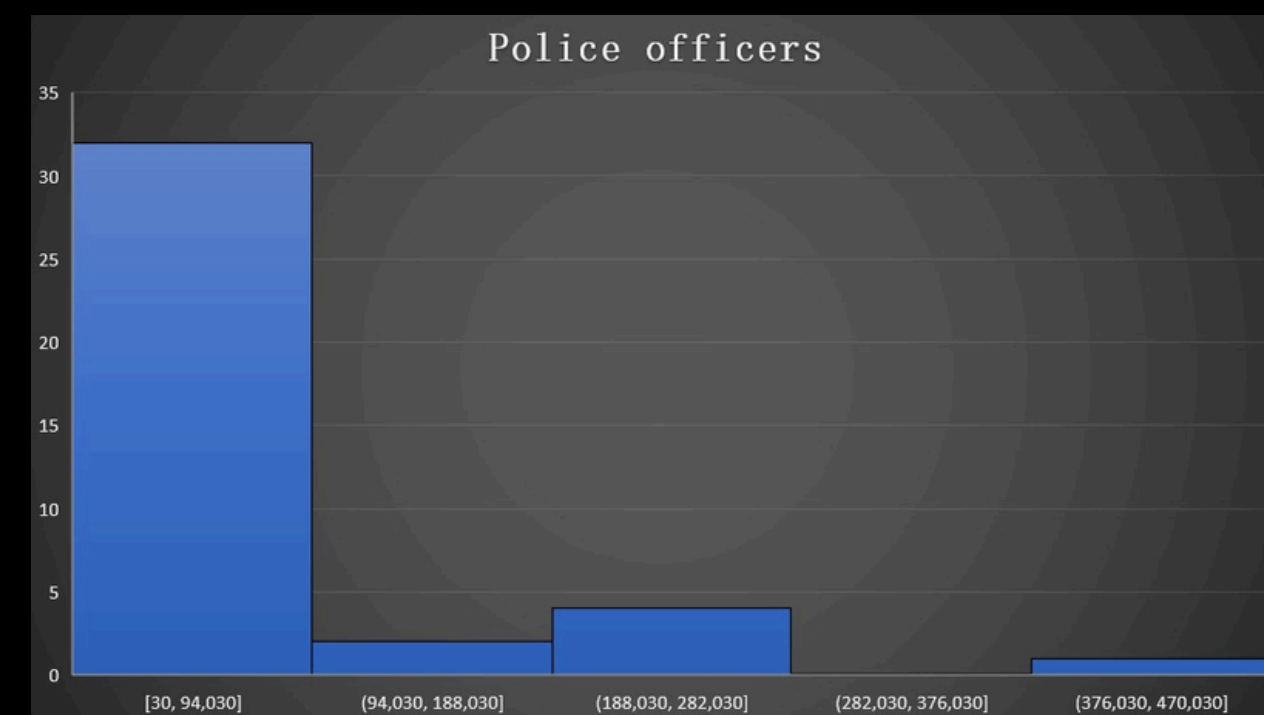
Suspects and offenders by citizenship



Intentional homicide (convicted)



Police officers



DESCRIPTIVE STATISTICS MEASURES

Legal cases processed		Explanations
mean	513504.1282	There are 513504.1282 legal cases processed per country
median	72209	Half of the countries have less than 72209 legal cases processed, while the other half has more
mode	#N/A	None of the values repeat
standard deviation	1233352.918	Since the standard deviation is greater than the mean, the data is further from the mean
kurtosis	12.69094982	Kurtosis>0, then the distribution is leptokurtic (thin distribution)
skewness	3.519467341	Skewness>1 - a high level of positive skewness
range	5964977	The difference between the highest and lowest value in the data set
min	26	The lowest value in the data set
max	5965003	The highest value in the data set
sum	20026661	The sum of all the values
count	39	The number of all the values
coefficient of variation	2.401836422	The coefficient is larger than 0.35, meaning large variation



Prisoners by citizenship		Explanations
Mean	22,588	On average, there are 22,588 prisoners by citizenship
Median	6,602	Half of the countries have less than 6,602 prisoners by citizenship, while the other half has more than 6,602
Mode	#N/A	None of the values repeat
Standard Deviation	46184.96987	Since the standard deviation is greater than the mean, the data is further from the mean.
Kurtosis	20.31336298	Kurtosis >0, then the distribution is leptokurtic (think distribution)
Skewness	4.129377165	Skewness >1 - a high level of positive skewness, meaning a right hand distribution
Range	264,779	The difference between the highest and lowest value in the data set
Minimum	63	The lowest value in the data set
Maximum	264,842	The highest value in the data set
Sum	880,933	The sum of all the values
Count	39	The number of all the values
Coefficient of variation	2.044666081	The coefficient > 0.35, meaning large variation

Police officers		Explanations
mean	56,883	There are 56,883 police officers per country
median	20,040	Half of the countries have less than 18,413 legal cases processed, while the other half has more
mode	#N/A	None of the values repeat
standard deviation	90566.51987	Since the standard deviation is greater than the mean, the data is further from the mean
kurtosis	4.983992628	Kurtosis>0, then the distribution is leptokurtic (thin distribution)
skewness	2.311919445	Skewness>1 - a high level of positive skewness
range	395,402	The difference between the highest and lowest value in the data set
min	120	The lowest value in the data set
max	395,522	The highest value in the data set
sum	2,218,433	The sum of all the values
count	39	The number of all the values
coefficient of var	1.592157291	The coefficient is larger than 0.35, meaning large variation



DESCRIPTIVE STATISTICS MEASURES

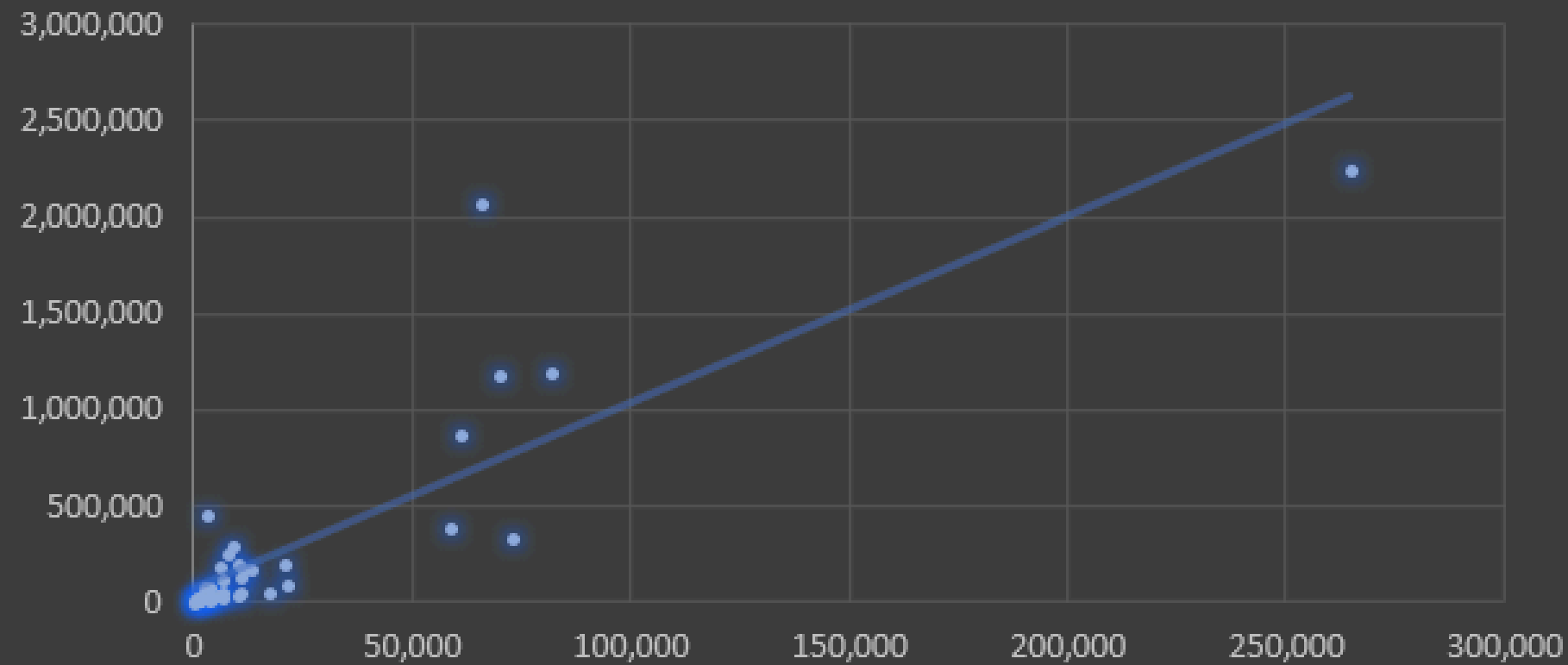


Intentional homicide(convicted persons)		Explanations
Mean	313	On average there are 285 290 suspects and offenders per country
Median	89	Half of the countries have less than 70 497 suspects and offenders, while the other half has more
Mode	114	There are 2 values of 114
Standard Deviation	1071.085305	Since the standard deviation is greater than the mean, the data is further from the mean.
Kurtosis	36.30709132	kurthosis>0, so the distribution is leptokurtic(thin distribution)
Skewness	5.940201146	Skewness>1 - a high level of positive skewness
Range	6,724	The difference between the highest and the lowest value
Minimum	0	The lowest value in the data set
Maximum	6,724	The highest value in the data set
Sum	12,214	The sum of values
Count	39	The number of values
Coefficient of variation	3.420036588	The coefficient is larger than 0.35, meaning large variation

Suspects and offenders by citizenship		Explanations
Mean	285,290	On average there are 285 290 suspects and offenders per country
Median	70,497	Half of the countries have less than 70 497 suspects and offenders, while the other half has more
Mode	#N/A	None of the values repeat
Standard Deviation	527435.5818	Since the standard deviation is greater than the mean, the data is further from the mean.
Kurtosis	7.295388757	kurthosis>0, so the distribution is leptokurtic(thin distribution)
Skewness	2.743673535	Skewness>1 - a high level of positive skewness
Range	2,246,378	The difference between the highest and the lowest value
Minimum	159	The lowest value in the data set
Maximum	2,246,537	The highest value in the data set
Sum	11,126,303	The sum of values
Count	39	The number of values
Coefficient of variation	1.848771123	The coefficient is larger than 0.35, meaning large variation



Relationship between prisoners
and
suspects & offenders

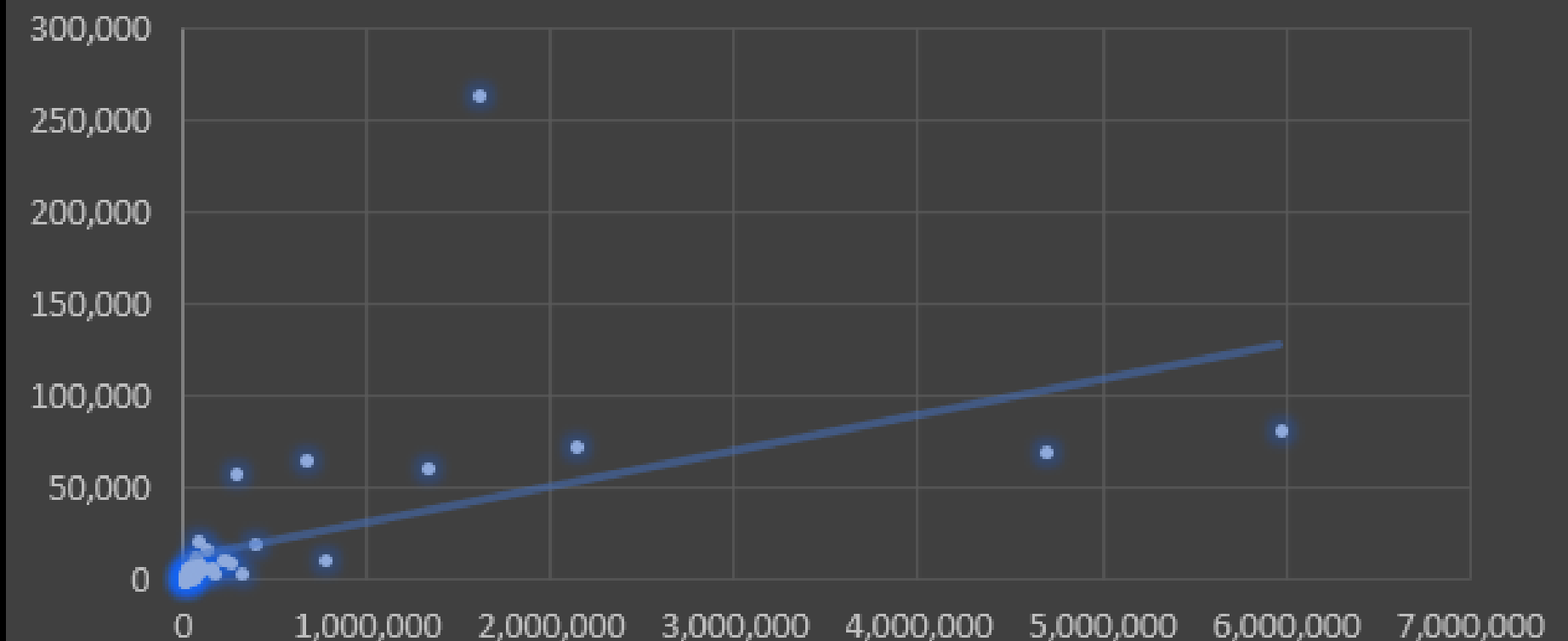


THE 2 VARIABLES LEGAL CASES
PROCESSED AND PRISONS
INFLUENCE RECIPROCALLY, HAVING
A STRONG CORRELATION DUE TO
THE VALUES' CLOSENESS TO THE
TREDLINE



THE 2 VARIABLES PRISONERS AND
SUSPECTS & OFFENDERS
INFLUENCE RECIPROCALLY, HAVING
A STRONG CORRELATION DUE TO
THE VALUES' CLOSENESS TO THE
TREDLINE

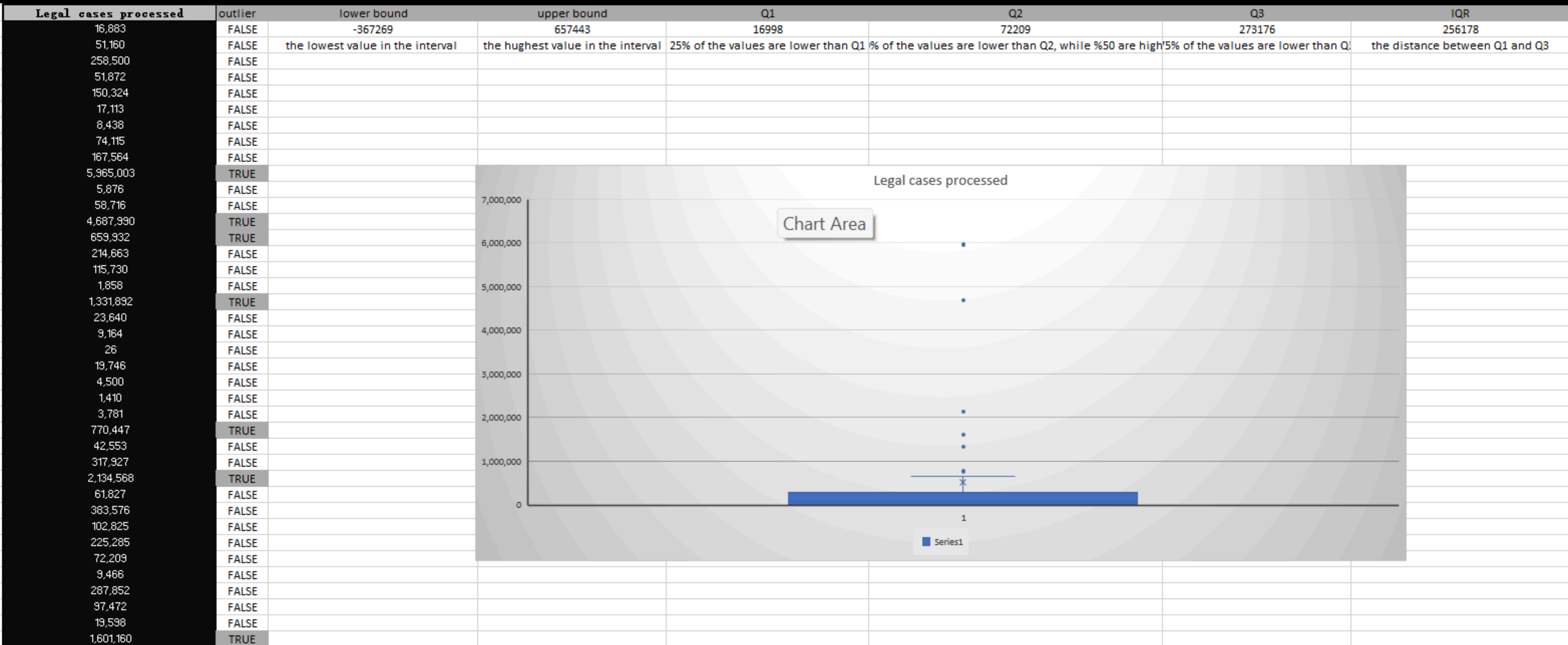
Relationship between legal cases processed
and
prisoners



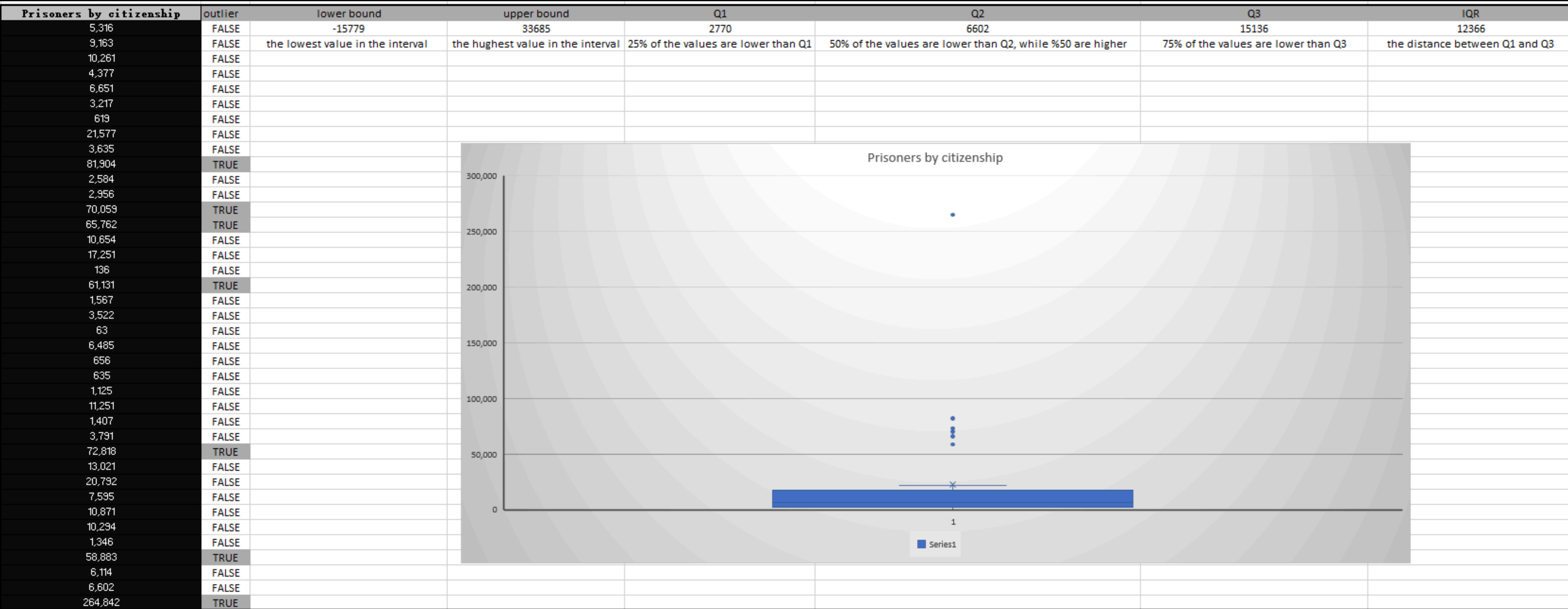
Covariance and coefficient of correlation

Legal cases-Prisoners by citizenship		Explanations
Covariance	29492043296	Covariance is greater than 0, meaning that the two variables tend to move in the same direction
Coefficient of correlation	0.517746096	Our coefficient is closer to 1, so there is a moderate to strong positive linear relationship
Prisoners by citizenship - Suspects and offenders by citizenship		Explanations
Covariance	20661145064	Covariance is greater than 0, meaning that the two variables tend to move in the same direction
Coefficient of correlation	0.848172715	Our coefficient is closer to 1, so there is a strong positive linear relationship
Suspects - Police officers		Explanations
Covariance	43926481083	Covariance is greater than 0, meaning that the two variables tend to move in the same direction
Coefficient of correlation	0.919579559	Our coefficient is closer to 1, so there is a positive and strong linear relationship
Police officers - Intentional homicide		Explanations
Covariance	68016204.02	Covariance is greater than 0, meaning that the two variables tend to move in the same direction
Coefficient of correlation	0.701165679	Our coefficient is closer to 1, so there is a moderate to strong positive linear relationship
Legal cases processed-Intentional homicide		Explanations
Covariance	333605976.2	Covariance is greater than 0, meaning that the two variables tend to move in the same direction
Coefficient of correlation	0.252535476	Our coefficient is closer to 0, so there is a weak linear relationship

BOXPLOT

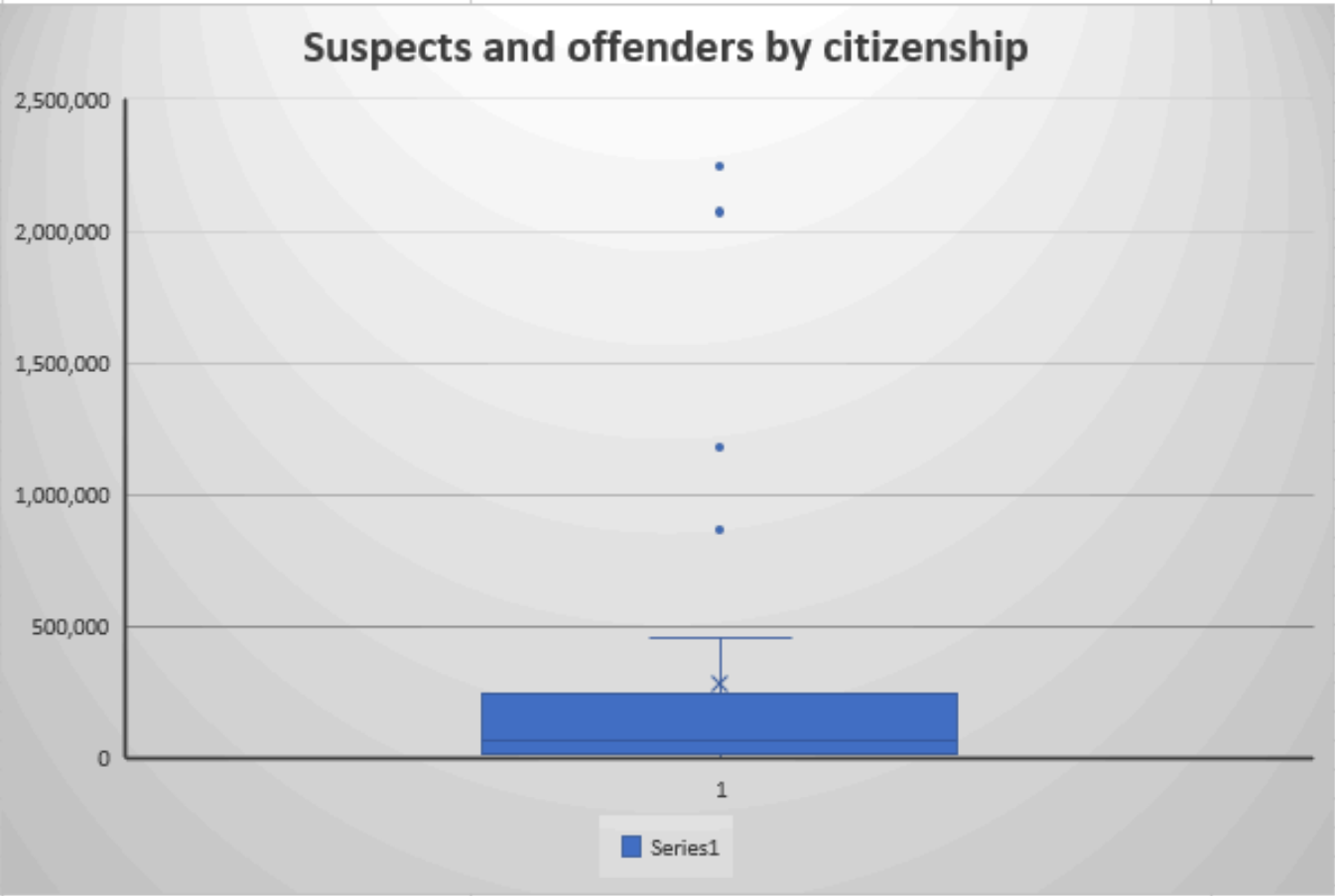


BOXPLOT



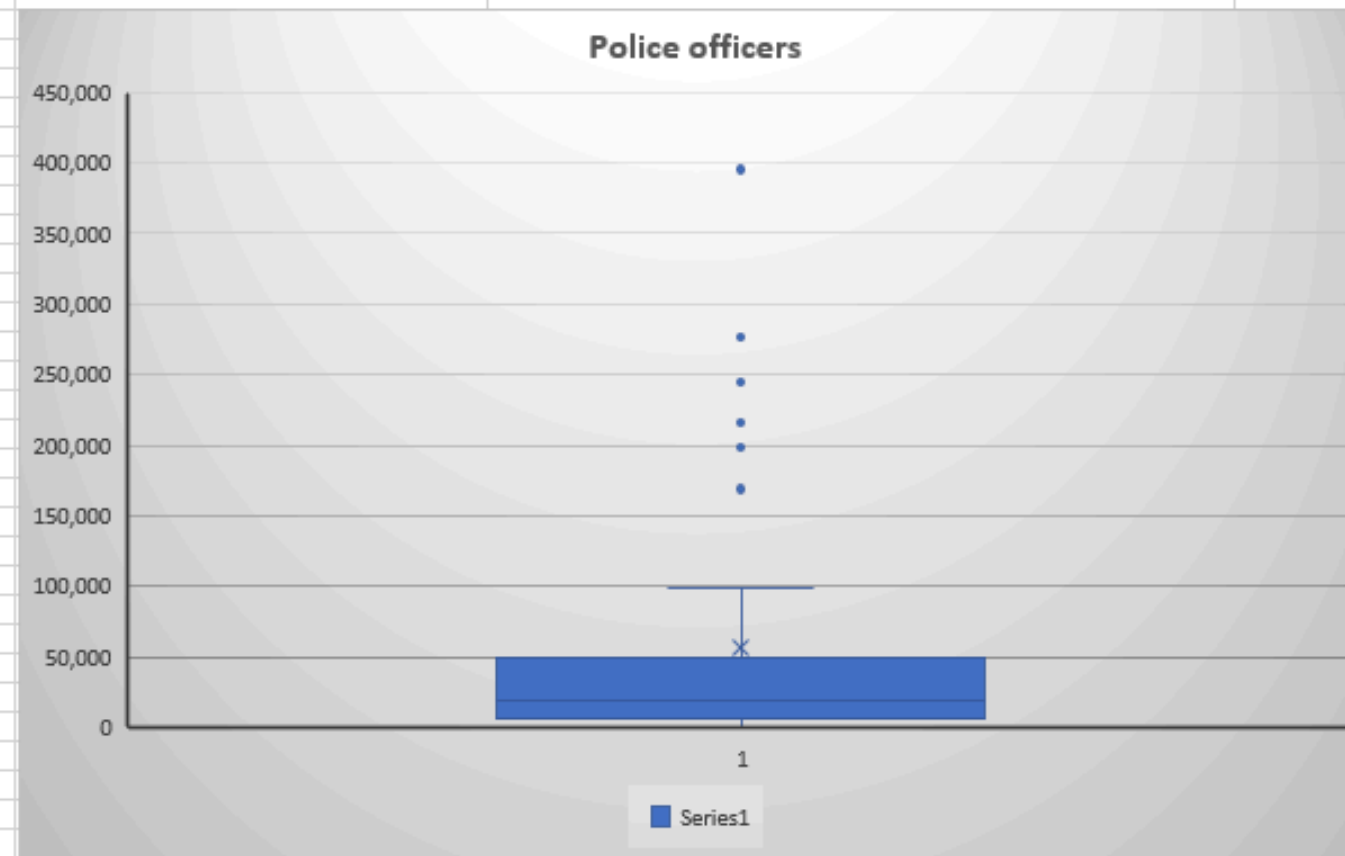
BOXPLOT

Suspects and offenders by citizenship	Outlier Check	Lower Bound	Upper Bound	Q1	Q2	Q3	IQR
39,649	FALSE	-277671.75	522694.25	22465.5	70497	222557	200091.5
288,414	FALSE	The lowest value in the interval	The highest value in the interval	25% of the values are lower than Q1	50% of the values are lower than Q2 while 50% are higher	75% of the values are lower than Q3	The distance between Q1 and Q3
198,634	FALSE						
19,216	FALSE						
48,449	FALSE						
17,399	FALSE						
4,347	FALSE						
84,620	FALSE						
34,042	FALSE						
1,190,000	TRUE						
75,191	FALSE						
454,726	FALSE						
1,178,414	TRUE						
2,070,987	TRUE						
124,257	FALSE						
52,475	FALSE						
6,596	FALSE						
869,304	TRUE						
34,450	FALSE						
13,384	FALSE						
159	FALSE						
24,075	FALSE						
29,245	FALSE						
5,791	FALSE						
4,425	FALSE						
165,960	FALSE						
20,856	FALSE						
70,497	FALSE						
336,886	FALSE						
174,822	FALSE						
196,461	FALSE						
246,480	FALSE						
50,278	FALSE						
41,426	FALSE						
15,579	FALSE						
389,149	FALSE						
180,770	FALSE						
122,353	FALSE						
2,246,537	TRUE						

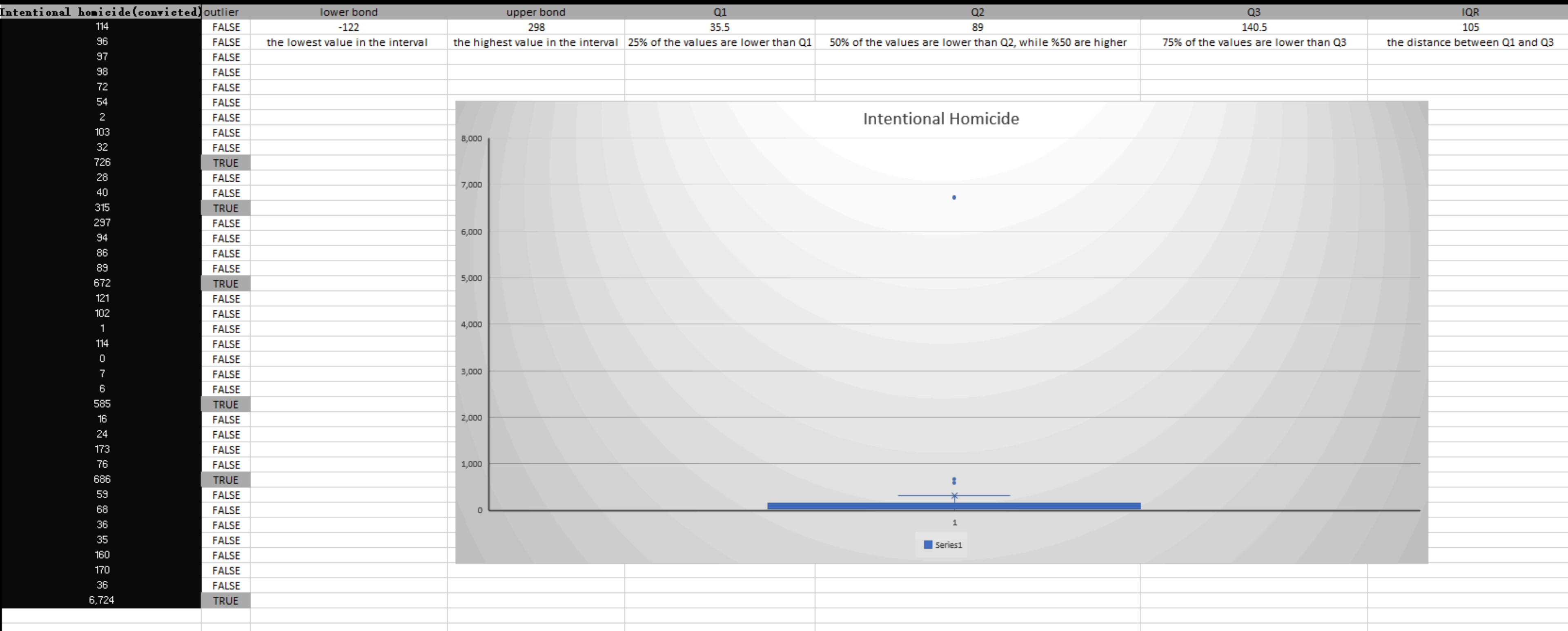


BOXPLOT

Police officers	Outlier Check	Lower Bound	Upper Bound	Q1	Q2	Q3	IQR
10,572	FALSE	-53082.75	108959.25	7683	20040	48193.5	40510.5
30,240	FALSE	The lowest value in the interval	The highest value in the interval	25% of the values are lower than Q1	50% of the values are lower than Q2 while 50% are higher	75% of the values are lower than Q3	The distance between Q1 and Q3
41,370	FALSE						
15,192	FALSE						
28,742	FALSE						
20,199	FALSE						
4,927	FALSE						
40,040	FALSE						
11,050	FALSE						
198,686	TRUE						
3,893	FALSE						
7,317	FALSE						
215,948	TRUE						
244,800	TRUE						
53,166	FALSE						
39,423	FALSE						
613	FALSE						
276,750	TRUE						
8,633	FALSE						
8,049	FALSE						
120	FALSE						
8,247	FALSE						
1,982	FALSE						
2,289	FALSE						
4,105	FALSE						
50,389	FALSE						
6,741	FALSE						
10,170	FALSE						
98,709	FALSE						
46,363	FALSE						
50,024	FALSE						
17,296	FALSE						
30,265	FALSE						
21,918	FALSE						
7,091	FALSE						
169,139	TRUE						
20,040	FALSE						
18,413	FALSE						
395,522	TRUE						



BOXPLOT



Spearman's coefficient

Country	Police Officers	Suspects	R _x	R _y	d _i ²
Austria	30240	288414	5	9	16
Belgium	41370	198634	8	8	0
Bulgaria	28742	48449	4	4	0
Croatia	20199	17399	2	1	1
Denmark	11050	34042	1	2	1
England and Wales	198686	1190000	10	10	0
Hungary	39423	52475	7	6	1
Romania	50024	196461	9	7	4
Serbia	30265	50278	6	5	1
Slovakia	21918	41426	3	3	0
Total	471917	2117578	55	55	24

Spearman's coefficient of ranking
0.854545455
formula: $1 - (6 \cdot 24) / 990$

Because it is closer to 1, the Spearman's coefficient demonstrates a positive and strong correlation between number of police officers and number of suspects.

Kendall's coefficient

Country	Police Officers	Suspects	R _x	R _y	concordant pairs(c)	discordant pairs(d)
Denmark	11050	288414	1	2	8	1
Croatia	20199	198634	2	1	8	0
Slovakia	21918	48449	3	3	7	0
Bulgaria	28742	17399	4	4	6	0
Austria	30240	34042	5	9	1	4
Serbia	30265	1190000	6	5	4	0
Hungary	39423	52475	7	6	3	0
Belgium	41370	196461	8	8	2	1
Romania	50024	50278	9	7	1	0
England and Wales	198686	41426	10	10	0	0
Total	471917	2117578	55	55	40	6

Kendall's coefficient of ranking
0.75555556
formula: $=2*(40-6)/90$

c = no. of units with ranks larger and following the current value
d = no. of units with ranks smaller and following the current value

Because it is closer to 1, the Kendall's coefficient demonstrates a positive and strong correlation between number of police officers and number of suspects.

DID YOU KNOW?

Iceland IS THE WORLD'S MOST PEACEFUL NATION, A
POSITION IT HAS HELD SINCE THE FIRST ITERATION OF THE
GLOBAL PEACE INDEX

The safest city is...

Vienna

IS A GREAT PLACE TO VISIT ACCORDING TO WORLD CAPITAL CITIES AND IT IS ONE OF THE TOP TEN SAFEST CAPITALS IN THE WORLD.

What about Bucharest?

2015-2020

AFTER A RESEARCH MADE BY STATISTA, WE KNOW THAT DISTRICT 5 RECORDED THE LOWEST CRIME RATE LEVEL IN BUCHAREST, TOTALING 62.98 PERCENT. BY CONTRAST, DISTRICT 3 HAD THE HIGHEST CRIME RATE, TOTALING 128.03 PERCENT.

Dangerous countries in Europe

IN EUROPE, THE BALTIC COUNTRIES HAD THE HIGHEST HOMICIDE RATES IN THE PAST YEARS. OF THESE, **LATVIA** HAD THE **HIGHEST** RATE, REACHING ALMOST 4.9 PER 100,000 INHABITANTS. MEANWHILE, THE **LOWEST** HOMICIDE RATE WAS FOUND IN **LUXEMBOURG**, AMOUNTING TO ONLY 0.32 HOMICIDES PER 100,000 INHABITANTS.

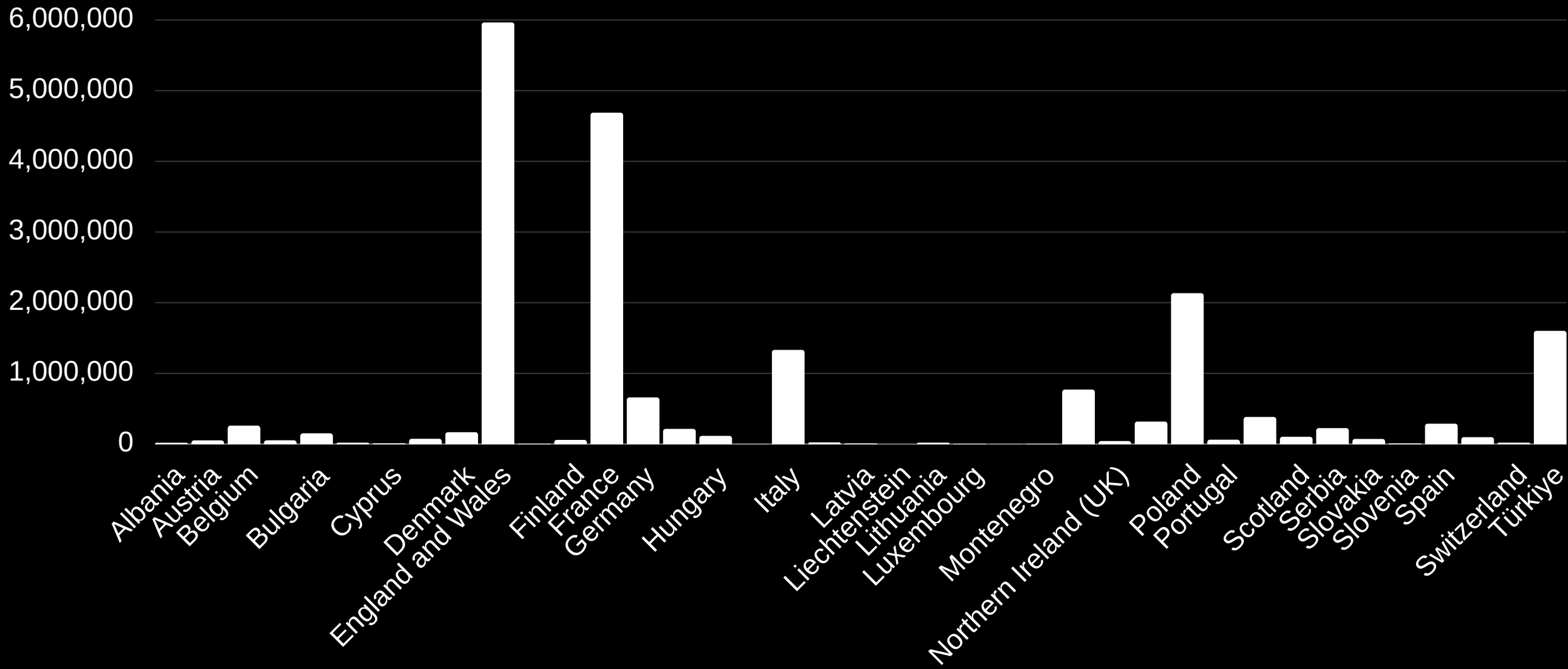
You are safe :)

NO. 22

ACCORDING TO THE GLOBAL PEACE INDEX, ROMANIA IS A PEACEFUL COUNTRY RANKING 22 WORLDWIDE. DESPITE EVERYTHING THAT IS GOING ON IN THE WORLD, ROMANIA REMAINS ONE OF THE SAFEST COUNTRIES IN CENTRAL AND EASTERN EUROPE, WITH A CRIME RATE BELOW THE EUROPEAN AVERAGE.

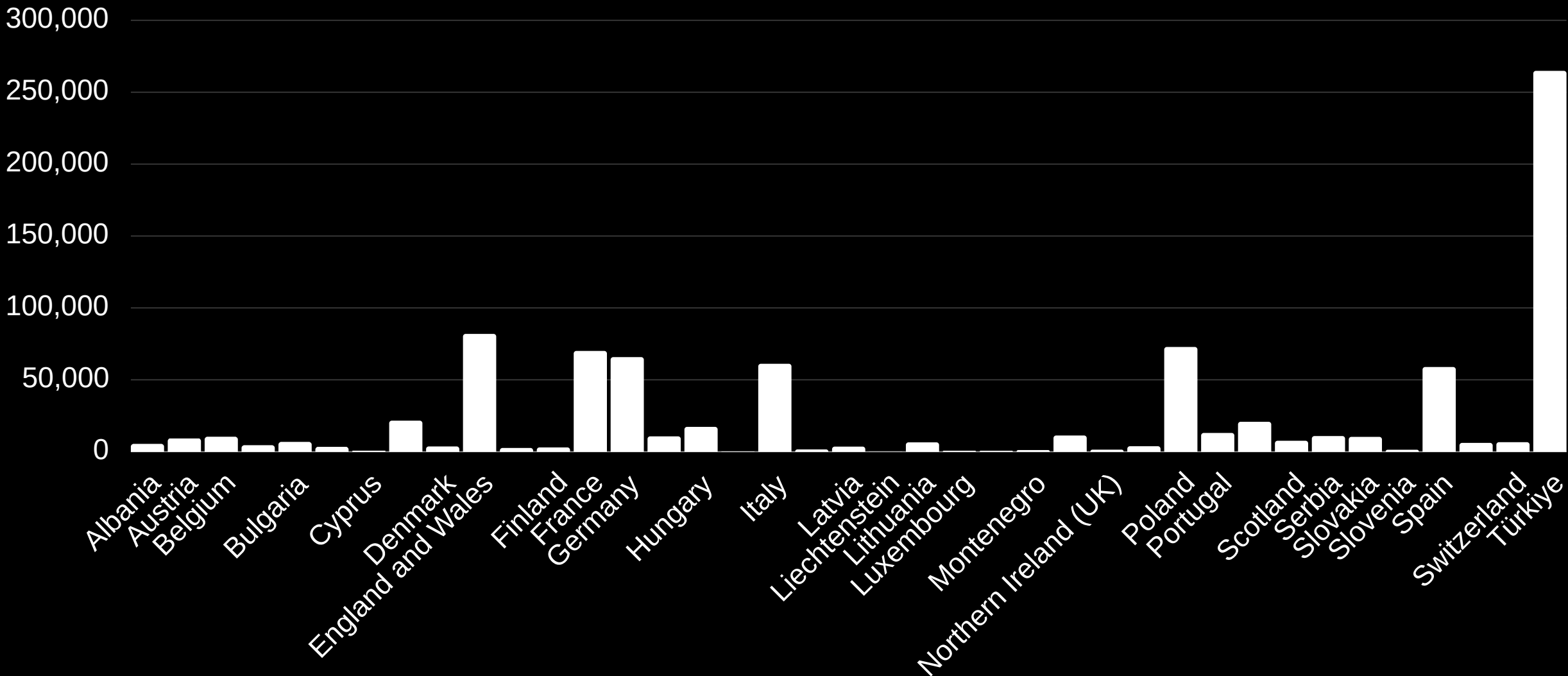
**Eurostat
data from
2018**

Legal cases processed in UE



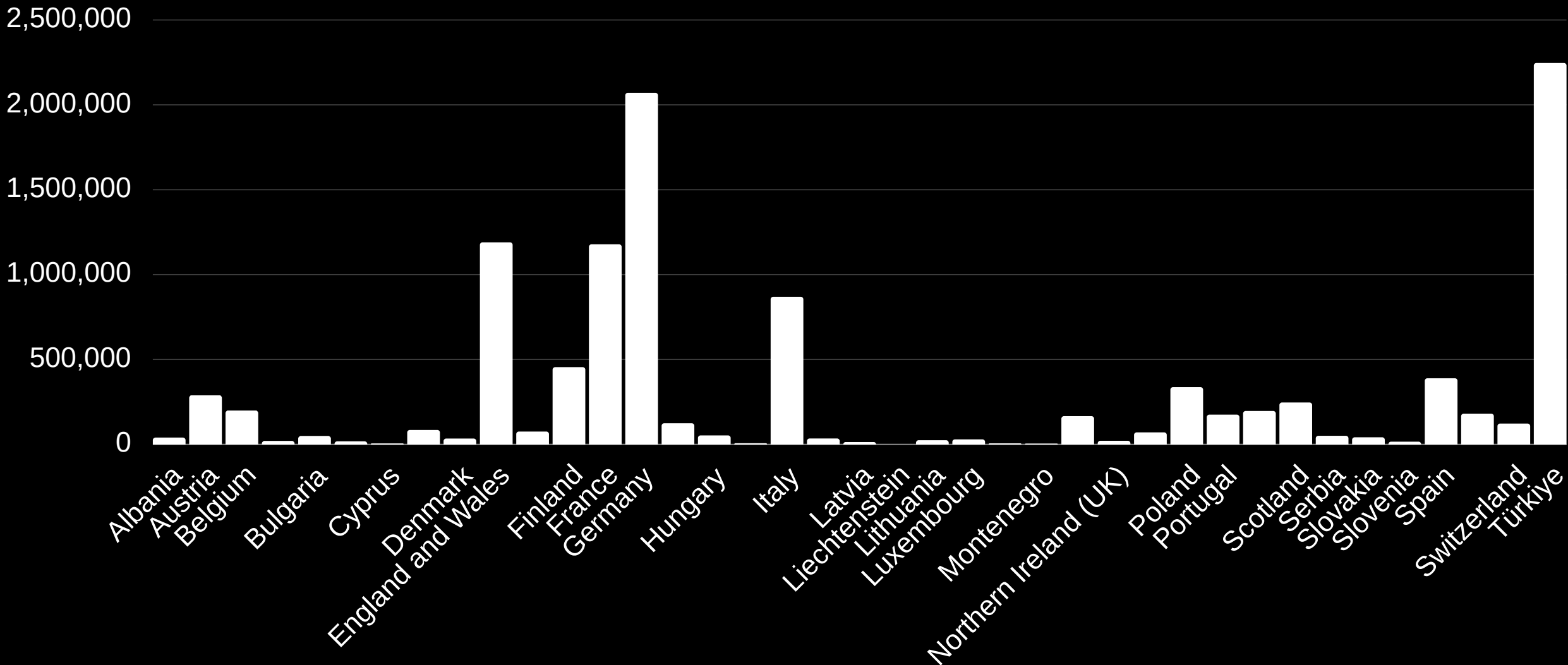
Eurostat
data from
2018

Total number of prisoners



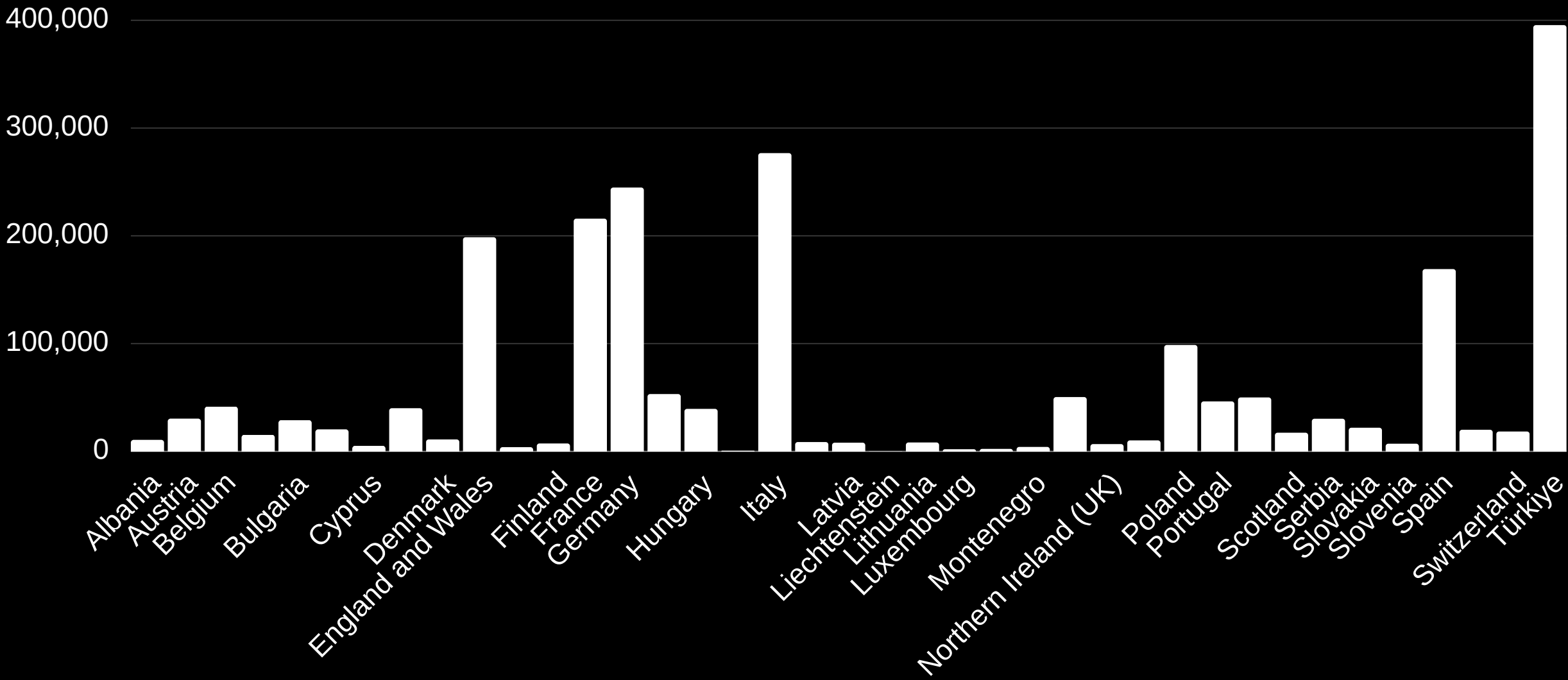
Eurostat
data from
2018

Total number of suspects



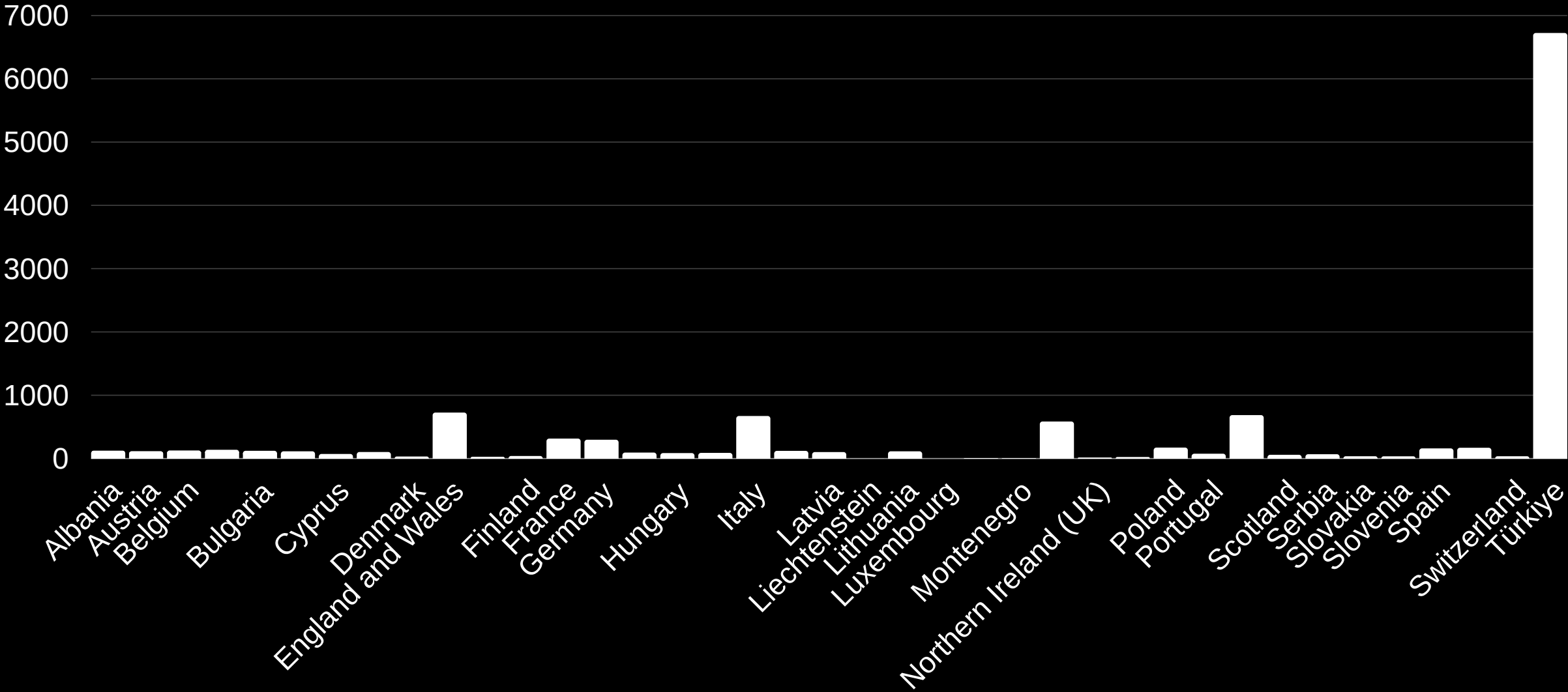
**Eurostat
data from
2018**

Number of police officers in UE



Eurostat
data from
2018

Crimes - intentional homicide



Thank you.